RAJALAKSHMI ENGINEERING COLLEGE

Mahalakshmi B 250701401-2025-CSE

M2

Started on	Thursday, 27 August 2026, 3:58 PM
State	Finished
Completed on	Thursday, 27 August 2026, 3:58 PM
Time taken	26 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

// Binary Search helper function (Divide and Conquer strategy)
int binarySearch(int arr[], int low, int high, int target) {
    if (low > high) {
        return -1;
    }

    int mid = low + (high - low) / 2;

    if (arr[mid] == target) {
        return mid;
    }
    if (arr[mid] > target) {
        return binarySearch(arr, low, mid - 1, target);
    }
    return binarySearch(arr, mid + 1, high, target);
}

// Recursive function to find the pair using Divide and Conquer binary search
int findPairRecursive(int arr[], int index, int n, int x) {
    if (index >= n) {
        return 0; // Pair not found
    }

    int complement = x - arr[index];

    // Search for the complement in the remainder of the array
    int pairIndex = binarySearch(arr, index + 1, n - 1, complement);

    if (pairIndex != -1) {
        printf("%d\n%d\n", arr[index], arr[pairIndex]);
        return 1; // Pair found
    }

    // Check the next element recursively
    return findPairRecursive(arr, index + 1, n, x);
}

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;

    int arr[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    int x;
    scanf("%d", &x);

    if (!findPairRecursive(arr, 0, n, x)) {
```

	Input	Expected	Got	
✔	4 2 4 8 10 14	4 10	4 10	✔
✔	5 2 4 6 8 10 100	No	No	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course